

Experience geology of Great Allegheny Passage

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Much has been written about the Great Allegheny Passage.

If you haven't heard, most of the multipurpose trail between Pittsburgh and Cumberland, Md., is completed, covering 153 miles. The path is for bicycling, hiking and walking, and thousands of people use it each year.

Bicycling is particularly popular, and an increasing number of cyclists are traversing the trail's entire length. Many trail users continue on to the 183-mile Chesapeake & Ohio Canal Towpath between Cumberland and Washington, D.C., making the journey about 330 miles. Most riders complete both trails in between five and seven days.

Consider a five-day trip. That comes out to about 66 miles each day. At a speed of 10 mph, you are riding for six hours a day. Add lunch stops and other breaks, and it makes for long, busy days that put the focus on cycling. At that pace, you get to enjoy some of the scenery from the seat of the bike.

In my opinion, however, you miss the marvelous details that the path has to offer.

Some of the more interesting details that you might miss along the Great Allegheny Passage are the exceptional array of geologic features. These features start at the beginning along the Potomac River at the edge of the Ridge and Valley Province in Cumberland and continue all the way to the Point in Pittsburgh, which is deep in the Pittsburgh plateau region.

One quickly notices the numerous rock outcrops. These are natural cliffs and rock outcrops or narrow cuts carved into hillsides and through small hills when the Western Maryland Railway was being built in 1911. One feature that you notice is that rock layers along the trail are angled.

The degree and direction varies. One place with angled rock layers is about a mile east of the Big Savage Tunnel and just before the McKenzie Hollow Road underpass. There, the layers are canted upward to the west. Tilted rocks are a result of rock layers under enormous pressure from one side, in this case pressure from the east, and being locked in place at the other side, to the west. Although we think of rock being solid, under the pressures that were involved over time, the layers actually folded like a stack of flexible paper pushed on one side and held at the other.

Just as the paper curves into waves, the rock layers bent into a series of bumps with low troughs between the higher bumps. What is exposed at this location, and others along the way, are cross-sections of the folds with wave peaks and

troughs.

Geologists call the raised parts of the folds anticlines. The troughs in between are called synclines. In the narrow rock cut just east of the McKenzie Hollow Road underpass, the tilted layers are a part of the eastern side of the Allegheny Mountain anticline. West of there, the Great Allegheny Passage cuts through the Negro Mountain, Laurel Ridge and Chestnut Ridge anticlines.

Just a few hundred yards farther northeast of the tilted rock is a concrete underpass below McKenzie Hollow Road and another important geologic or geographic feature of the Great Allegheny Passage. This is the Eastern Continental Divide.

Water falling on land to the east drains into small streams that join Wills Creek. That creek joins the Potomac River at Cumberland, then flows into the Chesapeake Bay and Atlantic Ocean.

To the west, water goes into Flaugherly Creek, the Casselman, Youghiogheny, Monongahela, Ohio and Mississippi rivers and, finally, into the Gulf of Mexico. A more precise way of looking at this watershed divide is as a sub-continental divide, because all the water ends up in the Atlantic Ocean. The continental divide along the Rocky Mountains is a true continental divide, because it splits water going into two different oceans.

The divide is the highest elevation of the Great Allegheny Passage, 2,392 feet above sea level. From there, the Passage winds north and west toward Connellsville, Fayette County. Over that route, it passes through Negro Mountain and Laurel and Chestnut ridges. These deep cuts through the mountains are an indication that the rivers are older than the mountains. These deep gorges are another geologic highlight called water gaps. The rugged, spectacular beauty of these three water gaps is one of the hallmarks of the Great Allegheny Passage.

Between Confluence and Ohiopyle, the Youghiogheny River cuts through Laurel Ridge. At the deepest, the difference between the top of the mountain and the level of the river is 1,662 feet, making this one of the deepest water gaps in Pennsylvania.

Between Ohiopyle and Connellsville, the tilted rocks of the west side of the Chestnut Ridge anticline are exposed alongside the trail. This is a good place to see the tilt of folded rock layers and layers of rocks in the region. Each layer is a different kind of rock derived from different environments.

Sandstones are common along the trail. The huge rocks in Ohiopyle State Park and the ledge over which Ohiopyle Falls drops and the boulders in the midst of all the rapids are Pottsville Sandstone. Sandstone was formed from sandy beaches of oceans that existed millions of years ago. As the landscape changed, the beaches were buried under other types of sediment. Pressure from the weight of the materials above and long periods of time compressed the sand into rock.

Limestone is a result of coral reefs and warm tropical oceans, where billions of

calcium-based animals, shelled creatures from microscopic to clam-sized, thrived. Where there were mud flats, shale layers formed. Finally, the teeming swamp forests of long ago, tall with tree ferns, are now layers of coal.

At Connellsville, the trail leaves the folded Allegheny Mountains behind and winds into the flat Pittsburgh plateau.

Below the surface of the plateau terrain are flat layered rocks. Just under the I-70 bridge in Westmoreland County, there are long layers of limestone, which highlight the flatness of the subsurface geology in this region. Again, other layers above or below might be sandstones, shale and coal.

Most of the geology along the Great Allegheny Passage is ancient. Most, but not all.

About a quarter-mile south of Cedar Creek Park are greenish-brown mounds of rock called tufa. This is the youngest rock you'll find, because it is still being formed. This is known as the Port Royal tufa. It is being formed as calcium from the ancient limestone is dissolved by slightly acidic groundwater. When that water comes back to the surface and evaporates, the calcium is left behind, and new layers of rock are being formed.

So, with a showcase of regional geology just a few feet off the Great Allegheny Passage, my advice is to slow down. Set your pace at a speed that allows you to take in all of the show.

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